Mindlin 板线性屈曲有限元分析

一、基本假设

板中面为 $x_3=0$，厚度 h 常数。

变量：

- 中面内位移：$u_\alpha(x_1,x_2)$，$\alpha=1,2$
- 横向挠度：$w(x_1,x_2)$
- 转角：$\phi_\alpha(x_1,x_2)$

位移场：

$$\bar{u}_\alpha(x_1,x_2,x_3)=u_\alpha(x_1,x_2)-x_3\phi_\alpha,\qquad \bar{u}_3=w$$

应变：

$$\epsilon_{\alpha\beta}=\varepsilon_{\alpha\beta}+x_3\kappa_{\alpha\beta}$$

中面应变与曲率：

$$\varepsilon_{\alpha\beta}=\frac{1}{2}(u_{\alpha,\beta}+u_{\beta,\alpha}),\qquad \kappa_{\alpha\beta}=-\frac{1}{2}(\phi_{\alpha,\beta}+\phi_{\beta,\alpha})$$

横向剪切应变：

$$\gamma_\alpha=2\epsilon_{\alpha3}=w_{,\alpha}-\phi_\alpha$$

二、本构关系

平面应力：$\sigma_{33}=0$。

$$\sigma_{\alpha\beta}=D_{\alpha\beta\gamma\eta}\,\epsilon_{\gamma\eta},$$

材料张量：

$$D_{\alpha\beta\gamma\eta}=\frac{E}{1-\nu^2}\left[\nu\,\delta_{\alpha\beta}\delta_{\gamma\eta}+\frac{1-\nu}{2}(\delta_{\alpha\gamma}\delta_{\beta\eta}+\delta_{\alpha\eta}\delta_{\beta\gamma})\right].$$

横向剪切：

$$\sigma_{\alpha3}=G\,\gamma_\alpha,\qquad G=\frac{E}{2(1+\nu)}.$$

三、弱形式

虚功:

$$\delta W_{\text{int}} = \delta W_{\text{ext}}.$$

三维内虚功：

$$\delta W_{\text{int}} = \int_V \sigma_{ij}\,\delta\varepsilon_{ij}\,dV = \int_V \big(\sigma_{\alpha\beta}\,\delta\varepsilon_{\alpha\beta} + \sigma_{\alpha 3}\,\delta\gamma_\alpha\big)dV.$$

代入

$$\delta\varepsilon_{\alpha\beta} = \delta\varepsilon^{(0)}_{\alpha\beta} + x_3\,\delta\kappa_{\alpha\beta}$$

$$\begin{aligned}
\delta W_{\text{int}} &= \int_V \sigma_{\alpha\beta}\left(\delta\varepsilon^{(0)}_{\alpha\beta} + x_3\,\delta\kappa_{\alpha\beta}\right)dV + \int_V \sigma_{\alpha 3}\,\delta\gamma_\alpha\,dV \\
&= \int_\Omega \left[\int_{-h/2}^{h/2} \sigma_{\alpha\beta}\,dx_3\right]\delta\varepsilon^{(0)}_{\alpha\beta}\,d\Omega + \int_\Omega \left[\int_{-h/2}^{h/2} \sigma_{\alpha\beta}\,x_3\,dx_3\right]\delta\kappa_{\alpha\beta}\,d\Omega + \int_\Omega \left[\int_{-h/2}^{h/2} \sigma_{\alpha 3}\,dx_3\right]\delta\gamma_\alpha\,d\Omega.
\end{aligned}$$

截面力：

- 膜力：$N_{\alpha\beta} = \int_{-h/2}^{h/2} \sigma_{\alpha\beta}\,dx_3$
- 弯矩:$M_{\alpha\beta} = \int_{-h/2}^{h/2} \sigma_{\alpha\beta}\,x_3\,dx_3$
- 剪力：$Q_\alpha = \int_{-h/2}^{h/2} \sigma_{\alpha 3}\,dx_3$

二維虚功形式

忽略膜力虚功项：

$$\delta W_{\text{int}} = \int_\Omega \Big(\underbrace{\int_{-h/2}^{h/2} \sigma_{\alpha\beta} x_3 dx_3}_{M_{\alpha\beta}} \delta\kappa_{\alpha\beta} + \underbrace{\int_{-h/2}^{h/2} \sigma_{\alpha 3} dx_3}_{Q_\alpha} \delta\gamma_\alpha\Big)d\Omega.$$

厚度积分，$k_s = \frac{5}{6}$：

$$\begin{aligned}
M_{\alpha\beta} &= D^b_{\alpha\beta\gamma\eta}\,\kappa_{\gamma\eta}, \qquad D^b_{\alpha\beta\gamma\eta} = \frac{h^3}{12} D_{\alpha\beta\gamma\eta}, \\
Q_\alpha &= S_{\alpha\beta}\,\gamma_\beta, \qquad S_{\alpha\beta} = k_s G h\,\delta_{\alpha\beta}.
\end{aligned}$$

线性双线性式:

$$\delta W_{\text{int}} = \int_\Omega \Big(\delta\kappa_{\alpha\beta} D^b_{\alpha\beta\gamma\eta}\,\kappa_{\gamma\eta} + \delta\gamma_\alpha S_{\alpha\beta}\,\gamma_\beta\Big)d\Omega.$$

四、几何刚度

4.1 实际载荷与载荷因子

压缩为正：

$$N_{\alpha\beta}^{0} = \lambda\, N_{\alpha\beta}^{\mathrm{ref}}.$$

λ 为载荷因子。

$N_{\alpha\beta}^{\mathrm{ref}}$ 为膜力合力；若给三维应力，先做厚度积分。

4.2 几何刚度的来源（非线性应变）

几何虛功：

$$\delta W_g = -\int_{\Omega} N_{\alpha\beta}^{0}\, w_{,\alpha}\, \delta w_{,\beta}\, d\Omega.$$

代入 $N_{\alpha\beta}^{0} = \lambda N_{\alpha\beta}^{\mathrm{ref}}$：

$$\delta W_g = -\lambda \int_{\Omega} N_{\alpha\beta}^{\mathrm{ref}}\, w_{,\alpha}\, \delta w_{,\beta}\, d\Omega.$$

4.3 屈曲时的总虛功平衡

总虛功：

$$\delta W_{\text{int,lin}} + \delta W_g = 0.$$

$$\int_{\Omega} \Big(\delta\kappa_{\alpha\beta} D^{b}_{\alpha\beta\gamma\eta}\, \kappa_{\gamma\eta} + \delta\gamma_{\alpha} S_{\alpha\beta}\, \gamma_{\beta}\Big) d\Omega - \lambda \int_{\Omega} N_{\alpha\beta}^{\mathrm{ref}}\ w_{,\alpha}\, \delta w_{,\beta}\, d\Omega = 0.$$

五、有限元離散（雙線性形式）

以節點基函數 $N_I(x_1, x_2)$ 離散。

5.1 位移場離散化

位移與轉角：

$$w_{,\alpha} = \sum_{I} N_{I,\alpha} w_I, \qquad \delta w_{,\beta} = \sum_{J} N_{J,\beta} \delta w_J$$

$$\phi_{\alpha,\beta} = \sum_{K} N_{K,\beta} \phi_{\alpha K}, \qquad \delta\phi_{\alpha,\beta} = \sum_{L} N_{L,\beta} \delta\phi_{\alpha L}$$

5.2 應變與曲率的離散表達

曲率與剪切應變：

$$\kappa_{\alpha\beta} = -\frac{1}{2}(\phi_{\alpha,\beta} + \phi_{\beta,\alpha}) = -\frac{1}{2}\sum_I (N_{I,\beta}\phi_{\alpha I} + N_{I,\alpha}\phi_{\beta I})$$

$$\gamma_\alpha = w_{,\alpha} - \phi_\alpha = \sum_I N_{I,\alpha} w_I - \sum_I N_I \phi_{\alpha I}$$

節點自由度：

$$\mathbf{d}_I = \begin{bmatrix} w_I \\ \phi_{1I} \\ \phi_{2I} \end{bmatrix}.$$

5.3 總體虛功平衡（離散形式）

$$\int_\Omega \Big(\delta\kappa_{\alpha\beta} D^b_{\alpha\beta\gamma\eta} \kappa_{\gamma\eta} + \delta\gamma_\alpha S_{\alpha\beta}\gamma_\beta\Big) d\Omega - \lambda \int_\Omega N^{\mathrm{ref}}_{\alpha\beta} w_{,\alpha} \delta w_{,\beta}\, d\Omega = 0$$

$$\boxed{a_b(\delta\phi, \phi) + a_s\big((\delta w, \delta\phi), (w, \phi)\big) - \lambda K_g(\delta w, w) = 0}$$

彎曲雙線性形式 a_b：

$$\begin{aligned} a_b(\delta\phi, \phi) &:= \int_\Omega \delta\kappa_{\alpha\beta} D^b_{\alpha\beta\gamma\eta} \kappa_{\gamma\eta}\, d\Omega \\ &= \sum_{IJ} \delta\boldsymbol{\phi}_J^T \mathbf{K}^b_{JI} \boldsymbol{\phi}_I \end{aligned}$$

$$\begin{aligned} a_b(\delta\phi, \phi) &= \int_\Omega \left[-\frac{1}{2}(\delta\phi_{\alpha,\beta} + \delta\phi_{\beta,\alpha})\right] D^b_{\alpha\beta\gamma\eta} \left[-\frac{1}{2}(\phi_{\gamma,\eta} + \phi_{\eta,\gamma})\right] d\Omega \\ &= \frac{1}{4}\int_\Omega (\delta\phi_{\alpha,\beta} + \delta\phi_{\beta,\alpha}) D^b_{\alpha\beta\gamma\eta} (\phi_{\gamma,\eta} + \phi_{\eta,\gamma}) d\Omega \\ &= \frac{1}{4}\sum_{I,J}\int_\Omega (N_{J,\beta}\delta\phi_{\alpha J} + N_{J,\alpha}\delta\phi_{\beta J}) D^b_{\alpha\beta\gamma\eta} (N_{I,\eta}\phi_{\gamma I} + N_{I,\gamma}\phi_{\eta I}) d\Omega \\ &= \sum_{I,J} \delta\boldsymbol{\phi}_J^T \mathbf{K}^b_{JI} \boldsymbol{\phi}_I \end{aligned}$$

矩陣形式：

$$\mathbf{K}^b_{JI} = \int_\Omega \left(\mathbf{B}^b_J\right)^T \mathbf{D}^b \mathbf{B}^b_I\, d\Omega$$

$\mathbf{B}^b_I$：

$$\left(\mathbf{B}^b_I \boldsymbol{\phi}_I\right)_{\alpha\beta} = -\frac{1}{2}(N_{I,\beta}\phi_{\alpha I} + N_{I,\alpha}\phi_{\beta I}).$$

剪切雙線性形式 a_s：

$$\begin{aligned} a_s\big((\delta w, \delta\phi), (w, \phi)\big) &:= \int_\Omega \delta\gamma_\alpha S_{\alpha\beta}\gamma_\beta\, d\Omega \\ &= \sum_{IJ} \delta\mathbf{d}_J^T \mathbf{K}^s_{JI} \mathbf{d}_I \end{aligned}$$

$$
\begin{aligned}
a_s\big((\delta w,\delta\phi),(w,\phi)\big) &= \int_\Omega (\delta w_{,\alpha}-\delta\phi_\alpha) S_{\alpha\beta}\,(w_{,\beta}-\phi_\beta)\,d\Omega \\
&= \int_\Omega \left[\left(\sum_J N_{J,\alpha}\delta w_J\right)-\left(\sum_J N_J\delta\phi_{\alpha J}\right)\right] S_{\alpha\beta}\left[\left(\sum_I N_{I,\beta}w_I\right)-\left(\sum_I N_I\phi_{\beta I}\right)\right] d\Omega \\
&= \sum_{I,J}\int_\Omega (N_{J,\alpha}\delta w_J - N_J\delta\phi_{\alpha J})S_{\alpha\beta}\,(N_{I,\beta}w_I - N_I\phi_{\beta I})d\Omega \\
&=: a_s^{ww}(\delta w,w) + a_s^{\phi\phi}(\delta\phi,\phi) + a_s^{w\phi}\big((\delta w,\delta\phi),(w,\phi)\big) \\
a_s^{ww}(\delta w,w) &= \sum_{I,J} S_{\alpha\beta}\int_\Omega N_{J,\alpha}\,N_{I,\beta}\,\delta w_J w_I\,d\Omega \\
a_s^{\phi\phi}(\delta\phi,\phi) &= \sum_{I,J}\int_\Omega N_J\,S_{\alpha\beta}\,N_I\,\delta\phi_{\alpha J}\phi_{\beta I}\,d\Omega \\
a_s^{w\phi}\big((\delta w,\delta\phi),(w,\phi)\big) &= -\sum_{I,J}\int_\Omega N_{J,\alpha}\,S_{\alpha\beta}\,N_I\,\delta w_J\phi_{\beta I}\,d\Omega \\
&\quad -\sum_{I,J}\int_\Omega N_J\,S_{\alpha\beta}\,N_{I,\beta}\,\delta\phi_{\alpha J}w_I\,d\Omega \\
&= \sum_{I,J}\delta\mathbf{d}_J^T\mathbf{K}_{JI}^s\mathbf{d}_I
\end{aligned}
$$

$$
\mathbf{K}_{JI}^s = \int_\Omega (\mathbf{B}_J^s)^T\mathbf{S}\mathbf{B}_I^s\,d\Omega
$$

$\mathbf{B}_I^s$：

$$
\mathbf{B}_I^s = \begin{bmatrix} N_{I,1} & -N_I & 0 \\ N_{I,2} & 0 & -N_I \end{bmatrix}, \qquad \mathbf{S} = k_s G h \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.
$$

幾何刚**度** K_g：

$$
\begin{aligned}
K_g(\delta w,w) &= \int_\Omega N_{\alpha\beta}^{\mathrm{ref}} w_{,\alpha}\delta w_{,\beta}\,d\Omega \\
&= \sum_{IJ}\int_\Omega N_{\alpha\beta}^{\mathrm{ref}} N_{I,\alpha}N_{J,\beta}w_I\delta w_J\,d\Omega
\end{aligned}
$$

5.4 幾何剛度（向量形式）

向量形式：

$$
K_g(\delta w,w) = \sum_{IJ}\delta\mathbf{w}_J^T\,\mathbf{K}_{JI}^g\,\mathbf{w}_I, \qquad \mathbf{K}_{JI}^g = \int_\Omega \left(\mathbf{B}_J^g\right)^T\mathbf{N}^{\mathrm{ref}}\mathbf{B}_I^g\,d\Omega
$$

$$
\mathbf{B}_I^g = \begin{bmatrix} N_{I,1} \\ N_{I,2} \end{bmatrix}, \qquad \mathbf{N}^{\mathrm{ref}} = \begin{bmatrix} N_{11}^{\mathrm{ref}} & N_{12}^{\mathrm{ref}} \\ N_{12}^{\mathrm{ref}} & N_{22}^{\mathrm{ref}} \end{bmatrix}
$$

總體離散方程

屈曲特徵值問題：

$$\boxed{\left[\mathbf{K}^b + \mathbf{K}^s - \lambda \mathbf{K}^g\right]\mathbf{d} = \mathbf{0}}$$

等價形式：

$$\boxed{\left(\mathbf{K}^b + \mathbf{K}^s\right)\mathbf{d} = \lambda \mathbf{K}^g \mathbf{d}}$$